

Maheer Ihtejaz Moin

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SUMMARY

Advanced Computing (Honours) student at the University of Sydney majoring in Computational Data Science, with hands-on experience in quantitative modelling, volatility forecasting, and statistical analysis in Python. Eager to apply data-driven problem-solving in a tech-driven trading environment.

EDUCATION

The University of Sydney Sydney, NSW 2024 – 2028

Bachelor of Advanced Computing (Honours), Dalyell Scholar, Vice Chancellor's Scholarship

Majors: Computational Data Science & Finance (Double Major)

WORK EXPERIENCE

2A Marketing Sydney | NSW (Mar 2023 – Dec 2023)

Business Analyst Intern

- Extracted, cleaned, and joined marketing campaign data using Python (Pandas) and built Seaborn visualisations to surface actionable insights, reducing manual reporting effort.
- Developed forecasting models to identify budget-reallocation opportunities, increasing return on investment (ROI) across key channels.
- Built and deployed a full-stack website (React, Node.js, Supabase) for product marketing and online sales, plus a shipment-tracking portal for shipments and documentation, hosted on the company domain.
- Gathered requirements directly from clients, translated technical data into plain-language insights, and tailored recommendations to their objectives.

PROJECTS

Future Volatility Forecasting (Optiver Industry Project)

- Engineered features from Optiver order-book data across 10 technology equities, computing realised volatility, log returns, bid-ask spread, weighted average price (WAP), and lagged volatility.
- Fitted a HAR-RV model on realised volatility to capture the long-memory structure of volatility.
- Built an ARMA-GARCH model on log returns to forecast future volatility for option-pricing applications.
- Trained an XGBoost regressor on the engineered predictors and lagged volatility, benchmarking it against the HAR-RV and ARMA-GARCH models.
- Built a trading engine that predicts the future volatility of a stock using all the fitted models for options pricing

Stock Portfolio Optimisation (Markowitz Mean-Variance Model)

- Implemented Modern Portfolio Theory (Markowitz mean-variance optimisation) in Python to build optimal portfolios from expected returns and the covariance matrix of historical equity prices.
- Applied quadratic optimisation (SciPy) to derive the efficient frontier and solve for the maximum-Sharpe-ratio and minimum-variance portfolios under weight constraints.
- Ran Monte Carlo simulations of thousands of random portfolios to validate the efficient frontier and visualise the risk-return trade-off.
- Computed annualised return, volatility, and Sharpe ratio per portfolio and visualised the efficient frontier, capital market line, and optimal allocations with Matplotlib.

End-to-End Automated Data Analysis Pipeline

- Built a fully automated pipeline from CSV upload to insight generation, cleaning data, running statistical tests, and generating charts with no manual steps.
- Wrote custom Python (Pandas, SciPy) to handle outliers and missing values and run inferential tests, including t-tests and Levene's test, across grouped data.
- Integrated the GPT-4 API to translate raw statistical results and visualisations into clear, human-readable summaries, delivered through an interactive Streamlit web app.

Efficient Email Filtering System with Bloom Filter

- Built a space-efficient email-filtering system using a Bloom filter for rapid, low-memory duplicate detection with zero false negatives, designed to scale to thousands of new entries per semester.
- Engineered a 7-hash Bloom filter over a 15,000-bit array with custom polynomial rolling hashes for fast membership checks under tight memory constraints.
- Implemented Base64 bit-array serialisation, CLI command parsing, and email ingestion from file and live input to support real-time ADD, CHECK, and SHOW operations.

TECHNICAL SKILLS

Languages: Python, Java, SQL, R, C++ **Web Development:** React, Node.js, Tailwind CSS

Data Science & Quantitative: NumPy, Pandas, SciPy, XGBoost, Matplotlib, Seaborn, Streamlit; statistical modelling, time-series forecasting (ARMA-GARCH, HAR-RV), hypothesis testing

Tools & Workflow: Git, GitHub, Gradle, Maven, Pytest, Docker, Kubernetes, GitHub Actions, Vercel, Supabase

